Weighted Automata Learning with a Hard Inductive Bias

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Main Idea

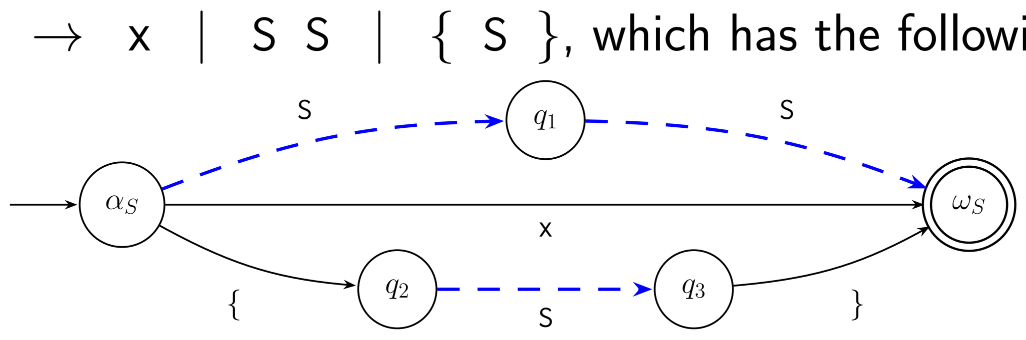
- **Assume**: we are given a learned distribution, $P_\theta(\sigma)$, with support over $\sigma : \Sigma^+$ but largely concentrated within $\mathcal{L}(G)$, where G is a known CFG.
- **Our goal**: distill a surrogate WFA, $\alpha_{\hat{\theta}}$, such that for all $\sigma \in \mathcal{L}(G)$, $P_{\hat{\theta}}(\sigma) \propto P_\theta\big(\sigma \mid \sigma \in \mathcal{L}(G)\big)$, while preserving inclusion $\big(\mathcal{L}(\alpha) \supseteq \mathcal{L}(G)\big)$.
- **Desiderata**: Regular description α should be reasonably concise and precise, i.e., $\Delta = |\mathcal{L}(\alpha) \setminus \mathcal{L}(G)|$ is approximately minimized up to Σ^n.

Nederhof's Regular Approximation

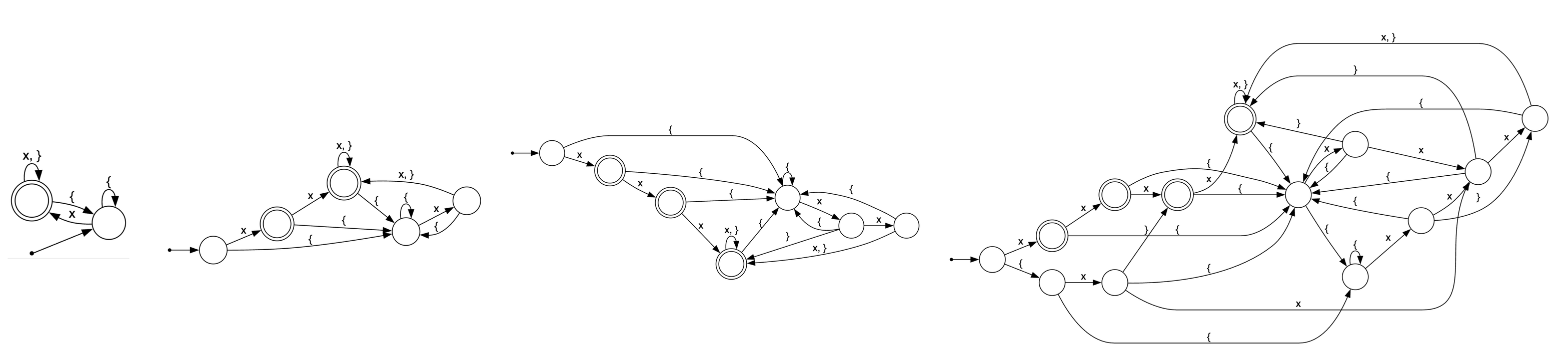
A recursive transition network (RTN) of a CFG $\langle \Sigma, V, P, S\rangle$ is an automaton,

$$\mathcal{R} = \big\{R_A = (Q_A, \delta_A, \alpha_A, F_A)\big\}_{A \in V}, \quad \delta_A \subseteq Q_A \times \big(\Sigma \cup V \cup \{\varepsilon\}\big) \times Q_A,$$

where an arc labelled $B \in V$ denotes a call to R_B. For example, consider the CFG, `S → x | S S | { S }`, which has the following RTN:

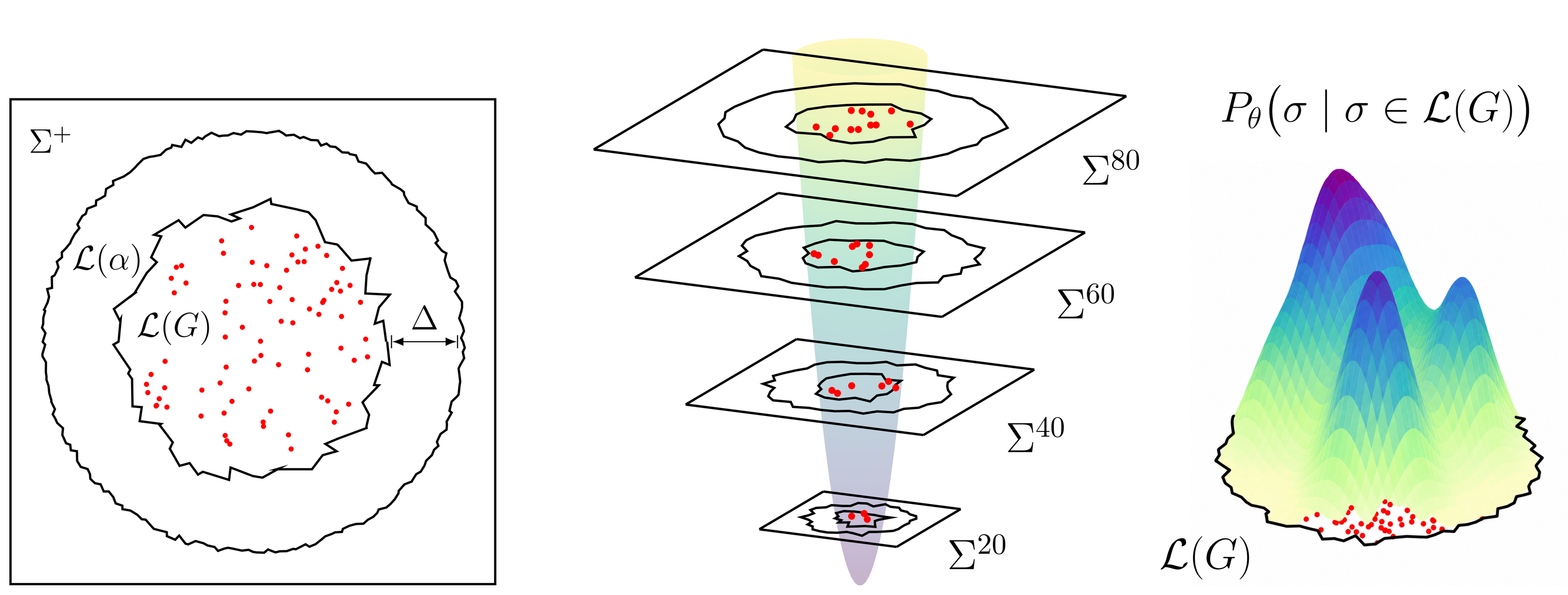


Nederhof's d-RTN approximation is a depth-bounded RTN unfolding. A nonterminal arc $p \xrightarrow{B} q$ is replaced by a fresh copy of the RTN for B, adding ε-arcs from p to the copy's initial state and the copy's final state back to q: $p \xrightarrow{\varepsilon} \alpha'_B,\ \omega'_B \xrightarrow{\varepsilon} q \quad (\forall \omega'_B \in F'_B)$. Unfold calls for d rounds. At depth d, every remaining nonterminal arc $p \xrightarrow{B} q$ is woven into a shared copy of R_B: $p \xrightarrow{\varepsilon} \alpha_B,\ \omega_B \xrightarrow{\varepsilon} q \quad (\forall \omega_B \in F_B)$. Below is the ε-removed, determinized and minimized DFA depicted for $1 \leq d \leq 4$:



$d = 1$ $\quad d = 2$ $\quad d = 3$ $\quad d = 4$

Language Overapproximation



We can measure over-approximation tightness by considering finite models,

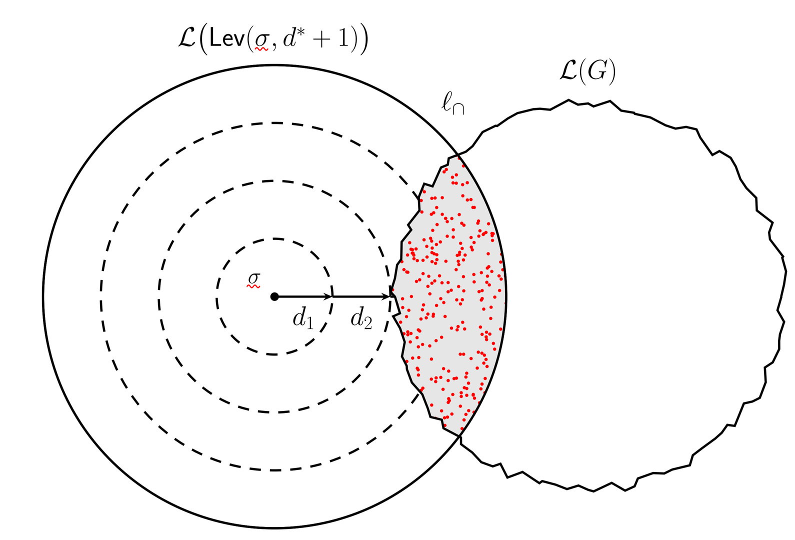
$$\rho(\alpha, G, m) = \frac{\sum_{n=1}^{m} |\mathcal{L}(G) \cap \Sigma^n|}{\sum_{n=1}^{m} |\mathcal{L}(\alpha) \cap \Sigma^n|}, \alpha \sqsubset_G^m \alpha' \iff \rho(\alpha', G, m) < \rho(\alpha, G, m).$$

Our Method

We propose a counterexample-guided abstraction refinement (CEGAR) loop that incrementally refines $\alpha_{\hat{\theta}}$ until a desired precision or $|Q|_{\max}$ is reached:

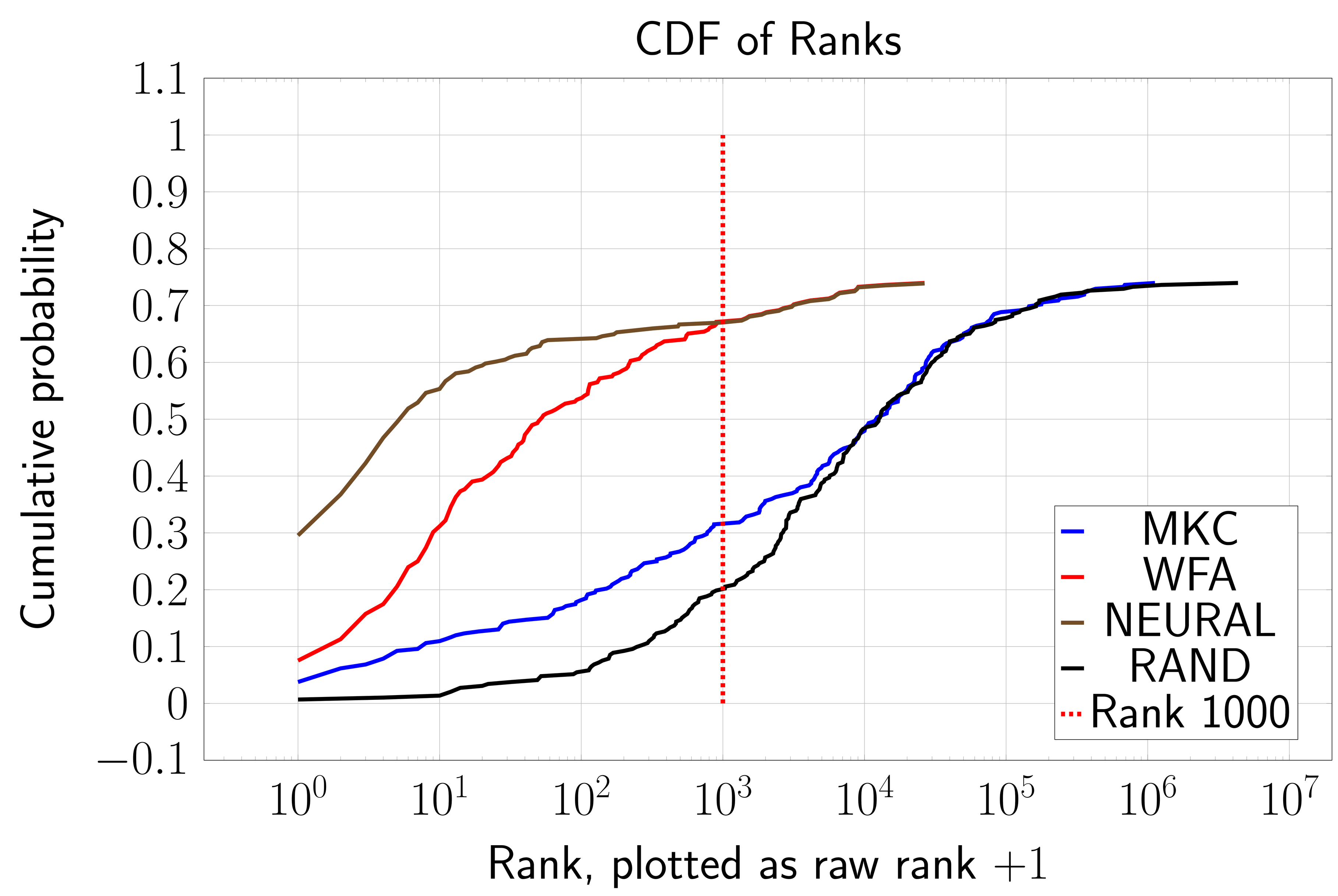
1. Initialize $\alpha'_{\hat{\theta}} \leftarrow \alpha_d$ using Nederhof's d-RTN construction.
2. Propose a refinement $D_{\mathsf{KL}}(P_\theta \| \alpha_{\hat{\theta}}) < D_{\mathsf{KL}}(P_\theta \| \alpha'_{\hat{\theta}})$ and $\alpha_{\hat{\theta}} \sqsubset_G^m \alpha'_{\hat{\theta}}$.
3. Binarize $\alpha_{\hat{\theta}}$ and verify $\mathcal{L}(\alpha) \supset \mathcal{L}(G)$ by checking $\varnothing = \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$.
4. If verification fails, collect $x \in \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$, otherwise go to step 2.
5. Repair α restoring inclusion, renormalize as $\alpha'_{\hat{\theta}}$, then go to step 2.

Maximum Probability Intersection Decoding



We will evaluate our learned WFA on a constrained decoding task: given a dataset of near-words $\underline{\sigma} \notin \mathcal{L}(G)$ and pairwise repairs σ within a small edit distance, sample uniformly WoR repairs within $\ell_\cap$, score and rank each repair according to $P_{\hat{\theta}}(\sigma)$, then measure mean reciprocal rank (MRR) of true repairs across the validation set. Optionally, we may rerank the top-1,000 repairs using a neural reranker, $P_\theta(\sigma \mid \underline{\sigma})$, and report MRR.

Evaluation



Takeaways & Next Steps

- A fixed-parameter $\langle k, |\mathbb{T}|\rangle$ tractable embedding from a syntax-directed type system to a CFG with soundness and completeness guarantees.
- Intersection with an acyclic FSA yields a star-free regular expression.
- Brzozowski derivatives enable incremental autoregressive decoding.
- Future work: lazy CNF materialization, quotienting (e.g., α-equivalence), richer type theories (subtyping, polymorphism, substructural constraints).